Rovem — Physics Model Specification

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Companion to the Software Requirements Specification, Draft 0.5. It defines the formulas, time step, control timings, measurement rules and automated tests. One model drives both the Build screen preview and the run (FR-3). A reference implementation of this specification reproduces all 135 values in the Simulation sheet of the Component Values sheet exactly, to the sheet's 3 decimal places.

Constants and units

All quantities are SI: metres, seconds, kilograms, newtons. Calculations use double-precision numbers and no random values (FR-29).

Constant	Value	Source
Gravity g	9.8 m/s²	SRS 2.2 and FR-29. Question text uses 10 m/s² (FR-19)
Time step Δt	1/120 s	Component Values sheet, Simulation
Throttle build-up T_t	1.0 s	Component Values sheet, Inputs
Brake build-up T_b	0.5 s	Component Values sheet, Inputs
Maximum brake force F_b,max	1500 N	Component Values sheet, Inputs (FR-6: fixed per rover model)
Rover length	2 m	Component Values sheet, Inputs

Part, surface, mission and power-up values come only from the Component Values sheet (NFR-5).

Build quantities

Each build is one chassis, one set of wheels and one motor (FR-1). The surface supplies μ_r (rolling friction) and μ_g (grip); the wheels supply the factors k_r and k_g (FR-9).

```
mass                m      = m_chassis + m_wheels + m_motor
weight              R      = m × g
rolling friction    F_r    = μ_r × k_r × R
grip limit          F_lim  = μ_g × k_g × R
max acceleration    a_max  = (min(F_d,max , F_lim) − F_r) ÷ m
max braking         b_max  = (min(F_b,max , F_lim) + F_r) ÷ m
top speed           v_max  = the motor's top speed
```

The Build screen shows these values for each surface in the mission (FR-2) and computes them with this same code (FR-3).

Controls

Throttle level u_t and brake level u_b run from 0 to 1 (FR-6).

- **Held:** the level rises by Δt ÷ T each step, up to 1. Throttle takes 1.0 s from 0 to 1, brake 0.5 s.
- **Released:** the level falls at the same rate, down to 0. The sheet's simulations hold the controls throughout, so no mission value depends on the release rate. It is a starting value for playtesting.
- **Both held:** the brake applies and the throttle counts as released (FR-6).
- **Level used in a step:** the level at the end of that step. From rest, the throttle level in step n is min(1, (n + 1) × Δt ÷ T_t).
- **Locks:** in Mission 2 the throttle drops to 0 at the power-cut line and stays locked (FR-39). In Mission 3 the throttle is locked for the whole attempt (FR-30).
- **Steering** changes the direction of travel, never the speed, and a stopped rover does not turn (FR-27). Distances are measured along the path driven. Turn rate: TBD, set in playtesting; it does not change any speed, distance or time in this specification.

Driving states and acceleration

In each step the rover is in exactly one state (SRS 3.4). Driving force is F_d = min(u_t × F_d,max, F_lim) and braking force is F_b = min(u_b × F_b,max, F_lim). The wheels slip, and the screen shows it, whenever u_t × F_d,max or u_b × F_b,max exceeds F_lim (FR-9).

State	When	Acceleration a
Stopped	v = 0 and either $F_d \leqslant F_r$ or the brake is held	0
Driving	Throttle applied, v below v_{max}	$(F_d - F_r) \div m$
At top speed	$v = v_{max}$ and $F_d \geqslant F_r$	0: the motor only balances friction (FR-28)
Cruising	Mission 3, before the tow is released	0: the tow line balances friction and holds the set speed
Coasting	No throttle and no brake while moving	$-F_r \div m$
Braking	Brake applied while moving	$-(F_b + F_r) \div m$, until v = 0

Braking and friction never push the rover backward: speed stops at 0 (FR-8).

The fixed time step, in order

The simulation advances in steps of $\Delta t = 1/120$ s, independent of the frame rate (FR-29). Within a step every force is constant, so speed and distance follow the equations of motion exactly (FR-10). Each step does the following, in this order:

1. Read the controls and update u_t and u_b (Controls).
2. Work out F_d, F_b and F_r from the current mass and surface.
3. Choose the state and its acceleration a (Driving states). If $v = v_{max}$ and $a > 0$, set $a = 0$.
4. Move for one step with that constant a:
   - Speed: $v' = v + a\Delta t$. Distance: $d = v\Delta t + \frac{1}{2}a\Delta t^2$.
   - If v' would pass v_{max}, the rover reaches v_{max} at $t_1 = (v_{max} - v) \div a$ and holds it for the rest of the step: $d = vt_1 + \frac{1}{2}at_1^2 + v_{max}(\Delta t - t_1)$.
   - If v' would fall below 0, the rover stops at $t_1 = -v \div a$ and stays stopped: $d = vt_1 + \frac{1}{2}at_1^2$.
5. Add d to the distance along the path and advance the clock by Δt.
6. Check the mission's lines and limits (Mission measurement).

While a power-up question is shown, no steps run and the mission timer stops (FR-33, FR-40).

Mission measurement and pass rules

Position x is the front of the rover, measured along the path from the start line. A line is reached at the end of the first step in which x reaches it, so times are multiples of Δt.

Mission	Start	Measured	Passed when
1. Motion Trial	From rest. Routes: 100 m on asphalt, or an 85 m shortcut on sand	Time to the target line	Time ⩽ 9 s
2. Surface Test	From rest on a 25 m ice strip, then sand	Ice time: time to the power-cut line at the end of the ice. Coasting distance: from x at the end of that step until the rover stops	Ice time ⩽ 8 s and coasting distance ⩾ 17.5 m
3. Stopping Challenge	On asphalt, towed at 15 m/s. The brake line is 23 m before the zone end; the zone is 6 m long	The first brake input after the brake line releases the tow and starts the brake build-up. Stopping distance runs from x at that input. Brake inputs before the line have no effect	The whole rover is inside the zone when stopped: zone start + 2 m ⩽ x ⩽ zone end

In Mission 3, releasing the brake after the tow is released lets the rover coast (FR-30). The tow speed of 15 m/s is above the Large motor's 14 m/s top speed, which is why the tow and not the motor holds the set speed.

Power-ups

A power-up multiplies one variable from the step after its question closes, for the rest of the attempt (FR-38). Only one is active; taking another replaces it (FR-40).

Power-up	Mission	Variable	Multiplier	Pickup position
Motor boost	1	F_d,max	× 1.2	Near the start of each route, beside Lighter load
Lighter load	1	m (so R, F_r and F_lim scale too)	× 0.9	Near the start of each route, beside Motor boost
Grip tyres	2	k_g	× 1.3	Start of the ice strip
Smooth bearings	2	k_r	× 0.7	End of the ice strip
Brake boost	3	F_b,max	× 1.2	One of two lanes before the brake line
Grip tyres	3	k_g	× 1.3	The other lane before the brake line

Motor boost raises F_d,max, but the driving force can never exceed F_lim. On sand it is capped by grip in 16 of the 27 builds, so part of the extra force is lost to wheel slip.

Checked against all 27 builds with this model:

- **Mission 1:** Taken at the start of the route, a power-up lets 5 of the 7 failing builds pass.
- **Mission 2:** Grip tyres let all 18 failing builds pass, and Smooth bearings let the 9 spiked-wheel builds pass.
- **Mission 3:** Brake boost lets all 4 failing builds pass. Grip tyres let none of them pass, because all 4 are limited by brake force, not grip.

Automated tests

Tests T1 to T9 compare the code with hand calculations at a relative error below 10^{-9} (NFR-4). T1 to T6 use the starting build: Standard chassis, All-terrain wheels, Medium motor.

Test	Setup	Expected
T1 Build quantities	Starting build	m = 209 kg, R = 2048.2 N. Asphalt: F_r = 40.964 N, F_lim = 1638.56 N. Sand: 245.784 N, 1024.1 N. Ice: 20.482 N, 204.82 N

Test	Setup	Expected
T2 Max acceleration	Starting build	Asphalt 4.588688995 m/s², sand 3.608688995 m/s², ice 0.882 m/s² (grip-limited)
T3 Max braking	Starting build	Asphalt 7.373033493 m/s², sand 6.076 m/s², ice 1.078 m/s²
T4 Moving off	Rest on asphalt, throttle held	Stopped in steps 0 to 3; moves first in step 4, where F_d = 41.667 N exceeds F_r = 40.964 N (FR-8)
T5 Exact steps	From rest at a = 4.588688995 m/s² for 120 steps	v = 4.588688995 m/s and x = 2.294344498 m, equal to at and ½at²
T6 Coasting	Sand, from 5 m/s, no controls	Stops after 10.629251701 m (v² ÷ 2a, with a = 1.176 m/s²)
T7 Top speed	At 16 m/s, throttle held	Speed stays 16 m/s; a = 0 (FR-28)
T8 Brake beats throttle	Both held while moving	Brake level rises, throttle level falls (FR-6)
T9 Frame-rate independence	Same input record at 30 and 144 frames per second	Identical position, speed and time after every step (FR-29)
T10 Sheet regression	All 27 builds, all three missions	All 135 Simulation sheet values, to 0.001 s or 0.001 m. Starting build: 8.500 s, 8.150 s, 7.642 s, 18.770 m, 18.767 m